Air Connectivity and International Travel: Evidence from Cross-border Card Payments*

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Abstract

Many countries seek to attract foreign travelers by improving air connectivity. How do direct flight connections affect international visitors' spending? We address this question using a novel dataset on card payments made by Chinese travelers via point-of-sale (POS) terminals. Our identification strategy exploits overseas improvements in air transportation capacity—arising from infrastructure developments, policy changes, and historical events—which we treat as exogenous supply shocks to flight frequency. Our IV estimates indicate that a 1% increase in the weekly frequency of direct flights leads to a 1.2% increase in cross-border card transaction value. While improving air connectivity promotes international travel, we find that negative shocks to consumer preferences for destination countries, such as boycotts, diminish the positive impact of air connectivity.

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JEL Classification: F10, F14, R41, L83

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1 Introduction

International tourism has grown steadily for over six decades, contributing 4.4% to the GDP of OECD countries (OECD 2020a). It ranks as the world's third-largest export category, following fuels and chemicals, and surpassing automotive products and food (UNWTO 2021). International visitors spend on food, shopping, and accommodations, which generates demand and creates jobs in destination countries (BEA 2023). Given its importance, policymakers actively pursue measures to increase inbound arrivals, including investments in air transportation infrastructure and the expansion of air connectivity.[1] However, limited evidence on how the development of air transportation promotes international travel hinders the evaluation of related policy initiatives. This paper addresses this gap in the literature and presents the first attempt to examine the effect of air connectivity on the spending of international visitors.

China is the largest spender on international travel, accounting for one-fifth of global travel spending followed by the US (UNWTO 2021). Our unique dataset contains Chinese consumer card transactions made in foreign countries through point-of-sale (POS) terminals. Unlike standard tourism statistics on the number of visitors, our dataset provides the amount of spending by Chinese travelers—a crucial factor for analyzing the economic impact of international travel. Transactions are observed between a given Chinese city (hereafter, origin city) and a foreign country (hereafter, destination country). We combine these transaction data with global flight schedules between origin cities and destination countries, which allows us to measure air connectivity between two locations. The resulting dataset is a year-origin city-destination country panel spanning 2011–16, where air connectivity is measured by the weekly frequency of direct flights.

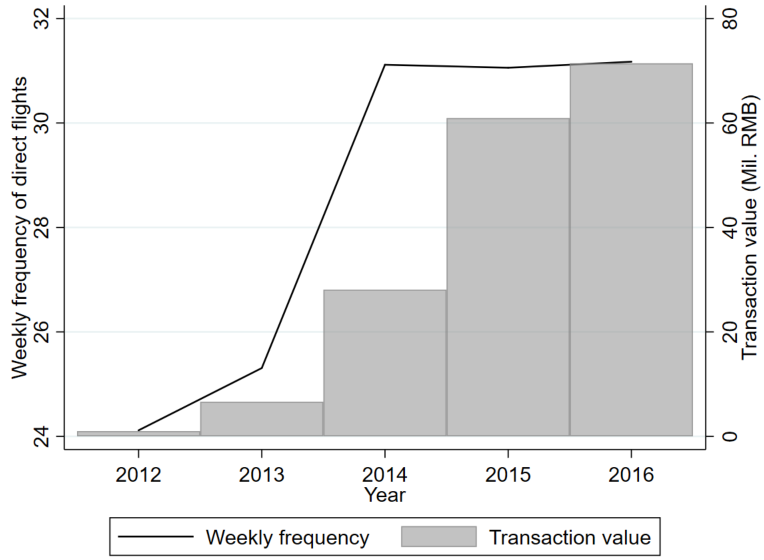
Our data present a clear picture of how direct flight connections relate to the spending of Chinese travelers. Figure 1 shows the weekly number of direct flights and the total value of card transactions from China to Qatar.[2] In Qatar, the Hamad International Airport started operations in April 2014 with a capacity twelve times that of the old airport.[3] In anticipation of the new airport opening, Qatar Airways launched new direct flight routes

[1] The importance of air transportation services in attracting foreign travelers has been widely discussed. For example, see OECD tourism papers: OECD (2018) and OECD (2020b).

[2] We focus on the Chinese cities connected with Qatar via direct flights. Prior to 2013, there were direct flight routes from Beijing, Chongqing, Guangzhou, and Shanghai to Doha. In 2013, Qatar Airways started operating three- and four-weekly flights from Chengdu and Hangzhou to Doha in September and December, respectively.

[3] Source for the information (link, accessed on July 6, 2025). This new airport would be able to accommodate 53 million passengers every year after all expansions.

Figure 1: Transaction Values and the Number of Direct Flights Between China and Qatar



Notes: We focus on the Chinese cities with direct flights to Qatar from 2012 to 2016. Transaction values are reported in millions of RMB.

from two Chinese cities (Chengdu and Hangzhou) to Doha in late 2013. These events added seven weekly direct flights from China to Qatar in 2013, and we observe a subsequent increase in the spending of Chinese travelers in Qatar.

We identify the effect of air connectivity on international travel using a gravity equation with three-way fixed effects (FEs): Chinese city-foreign country, Chinese city-year, and foreign country-year FEs. However, the OLS estimate may be biased due to measurement error or heterogeneous treatment effects. To address this concern, we instrument for air connectivity using overseas developments in air transportation as exogenous shocks. Specifically, our instrument interacts the global share of total flights departing from a destination country (capturing that country's comparative advantage in air transportation) with the geographic distance between a Chinese city and a foreign country (distant locations are less connected by flights; Cristea 2023). The interaction captures the differential impact of improvements in air transportation capacity on travelers in proximate versus distant cities.

Our identification strategy relies on the assumption that overseas developments in flight capacity are uncorrelated with demand shocks from Chinese travelers to those countries. We document that these developments were driven primarily by supply-side factors, including airport expansion projects by foreign governments and historical events. Moreover, we show that although destination countries with improved flight capacity receive more inbound flights overall, the share of flights originating from China remains stable. This supports our identification assumption: the objective of expanding air transportation capacity is to attract more travelers from around the world, rather than specifically targeting travelers from China. Further, the inclusion of three-way fixed effects ensures that identification comes from variation across destination countries, origin cities, and years. For example, country-year FEs control for time-varying destination characteristics, such as changes in visa requirements or improvements in attractiveness as a travel destination.[4]

Our IV estimate indicates that a 1% increase in the weekly frequency of direct flights leads to a 1.2% increase in the value of card transactions in the destination country. This result suggests that if a city with the average number of direct flights gains one additional monthly direct flight, travelers make 49.18% more card payments in that destination country. We also find similar evidence using travel time as an alternative measure of air connectivity: reduced travel time is associated with a sizable increase in overseas spending. Our study informs policy formulation, especially in assessing the impact of investments in air transportation infrastructure on traveler spending. We use the opening of the new Istanbul Airport in Turkey to illustrate how such infrastructure investments can generate considerable visitor spending relative to their cost.

Furthermore, improvements in air connectivity affect not only travelers' spending (i.e., imports of tourism-related services and goods) but also trade in goods, particularly imports of consumer products. Passenger planes often carry air freight, and exposure to new products abroad may shape consumer preferences and increase imports. Using customs data, we find that more frequent direct flights increase imports of consumer goods but have no effect on non-consumer goods. We also examine political conflicts as exogenous shocks to the attractiveness of travel destinations. Our results show that negative sentiment toward a destination country reduces the positive effect of air connectivity on travel—fewer Chinese consumers take advantage of direct air connections when public sentiment shifts against destination countries. However, for non-consumer goods imports, political conflicts do not affect the impact of air connectivity. This contrast suggests that international travel is more sensitive to consumer preferences and perceptions than goods trade, particularly in the case of non-consumer goods.

This paper contributes to several strands of the literature examining the effects of international air transportation on economic development (Hovhannisyan and Keller 2015; Campante and Yanagizawa-Drott 2018; Cristea 2023), international trade (Cristea 2011; Alderighi and Gaggero 2017; Wang et al. 2025; Söderlund 2023), foreign investment (Cam-

[4]Additionally, origin city-destination country FEs ensure that distance (a component of our IV) affects the dependent variable (card transaction values) only through air connectivity.

pante and Yanagizawa-Drott 2018; Fageda 2017; Tanaka 2019), and cross-border mergers and acquisitions (Zhang et al. 2021). While attracting foreign travelers is a key policy goal, little is known about how improvements in air transportation networks affect international traveler spending. We advance this literature by examining the causal impact of air connectivity on cross-border travel and contrasting it with goods trade. To address endogeneity concerns, we use overseas expansions in air transportation capacity as exogenous shocks to the development of direct flights. This approach builds on Cristea (2023), who uses proximity to the domestic aviation hubs, and Söderlund (2023), who exploits the liberalization of the Soviet airspace.

We also relate to the literature on cross-border travel. Prior studies focus on consumer demand for cross-border shopping between Sweden and Denmark for specific products, such as alcohol (Asplund et al. 2007) and groceries (Friberg et al. 2022). Chandra et al. (2014) and Baggs et al. (2018) investigate how travelers respond to exchange rate fluctuations in cross-border travel between Canada and the US.[5] These studies focus on proximate countries and highlight the roles of price differentials and travel costs (proxied by distance). In contrast, our analysis considers international travel between non-contiguous countries, which is increasingly common as air transportation becomes more affordable. Our setting also covers a wide range of destination countries, allowing us to study how consumer sentiment toward a destination affects international travel.

Finally, our study contributes to the growing literature on the impact of tourism on local economic development. For example, Faber and Gaubert (2019) use Mexican micro-level data and a spatial equilibrium model to show that tourism promotes local economic gains through its positive spillovers to manufacturing. Allen et al. (2021) combine card transactions data with payroll information in Barcelona, and find that tourism imposes a greater welfare loss on residents in the city center, who face a higher price despite higher wages. Among this line of research, Nocito et al. (2023) is the most closely related to our study. The authors exploit the differential timing of international releases of a TV series filmed in Sicily, Italy, using it as an exogenous shock to local tourism. Unlike these papers, we focus on overseas developments in air transportation capacity and examine how direct flight connections shape traveler spending in destination countries.

The outline of the paper is as follows. Section 2 introduces the data and presents stylized

[5]For example, Chandra et al. (2014) find that a stronger Canadian dollar against the US dollar (i.e., a lower foreign price for Canadians) motivates cross-border travel, and the response is mitigated by distance to the border. Baggs et al. (2018) find similar results but also show how Canadians' cross-border shopping negatively affects domestic retailers.

facts. Section 3 describes the empirical strategy. The main results using cross-border card transaction data are presented in Section 4, and Section 5 provides further analysis using customs data. Section 6 concludes.

2 Data and Stylized Facts

We use a unique dataset of cross-border card transactions made by Chinese travelers. We merge the card transaction data with international flight schedules to examine the impact of air connectivity on Chinese overseas travel spending. Our novel data show that China experienced a significant expansion in its air transport network, which is positively correlated with the value of cross-border card transactions.

2.1 Data Sources

(i) Chinese overseas card transactions

A unique dataset of Chinese on-site card transactions enables us to analyze spending by Chinese travelers abroad. The dataset covers the period from 2011 to 2016, obtained from a major Chinese consumer card provider.[6] The data include point-of-sale (POS) transactions made outside China by Chinese cardholders, excluding online transactions. All cards in the dataset are issued by domestic Chinese brands, excluding foreign brands such as Visa and MasterCard. This limitation is unlikely to bias our analysis, as the vast majority of Chinese consumers use domestic cards (e.g., 99% of total card payments in China).[7] Additionally, regulations limit travelers departing China to carrying no more than 3,100 USD in RMB, or 5,000 USD in foreign currency, without special permits. This restriction encourages them to rely more on card payments abroad rather than cash.[8]

We observe transaction values by cardholders' city of residence and the countries in which the transactions were made.[9] For confidentiality reasons, our data provider aggregates the information at the city of residence-destination country-year level. Although we do not

[6] Unfortunately, we only have data until August 2016 due to a regulatory change. We impute transaction values for the remaining months (September to December 2016) using monthly patterns from 2015 to account for seasonal variation in travel demand across regions. See Appendix B for the details.

[7] Reference: *Worldpay from FIS, The Global Payments Report 2023*, page 48

[8] Popular mobile payment platforms, such as WeChat Pay and Alipay, were not widely available outside China during our sample period. Even in Japan—the most popular destination for Chinese travelers—these services only became available later.

[9] The data provider imputes the city of residence based on the location with the highest number of past transactions. We exclude travelers whose activity suggests residence outside China (i.e., those with the most transactions in foreign countries).

Table 1: Summary Statistics

Variables	Mean	P(50)	P(5)	P(95)	SD	Observations
Card transactions						
Value (millions RMB)	14.17	0.19	0.001	38.31	134.10	42,940
Flight schedules						
Weekly frequency of direct flights	0.61	0	0	0.06	6.69	42,940
Travel time (hours)	12.75	13.44	5.60	19.77	4.63	34,584
Imports						
Value (millions RMB)	117.73	0.06	0	289.13	977.51	42,940

Notes: This table reports the mean, median (P50), 5th percentile (P5), 95th percentile (P95), standard deviation (SD), and the number of observations for each variable used in the regressions. The number of observations for travel time is smaller because no flight routes could be constructed for some city–country pairs, even allowing up to three stops.

observe transaction categories, the dataset offers valuable insights into total spending by Chinese consumers traveling from a given city to a given country each year from 2011 to 2016. The dataset includes 336 Chinese cities and 71 destination countries in Europe and Asia, which together account for 64% of global exports in travel services (excluding exports from China).[10]

(ii) Global flight schedules

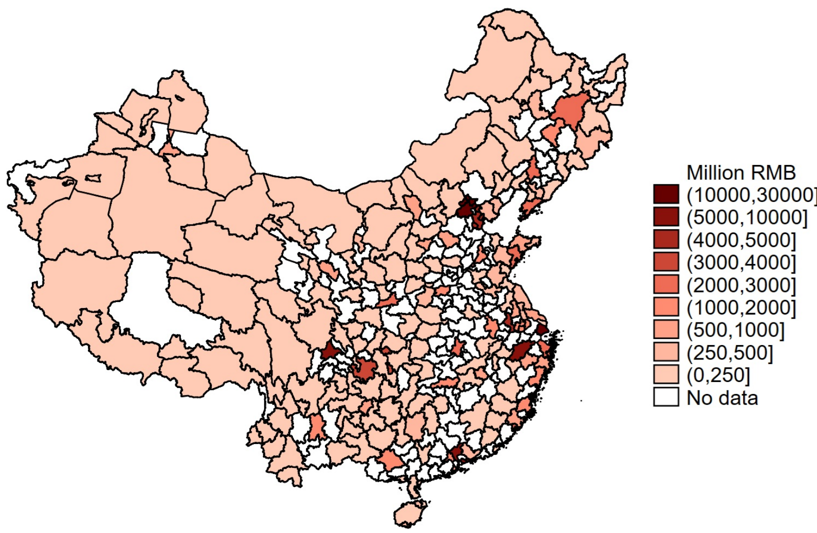
We obtain international flight schedule data from OAG Analyser, which provides worldwide flight schedules from 2011 to 2016. The dataset includes information on departure and arrival airports, scheduled times, elapsed time, travel distance, and number of stops. We augment the data with the names of Chinese cities served by each airport, along with the geographic coordinates of both Chinese cities and foreign countries. Our primary measure of air connectivity is the weekly frequency of direct flights between a Chinese city and a foreign country. We also construct an alternative measure based on travel time, which incorporates indirect routes. Specifically, we identify all feasible flight paths with up to two domestic stops and one international stop, adding a three-hour layover for each. We then select the route with the shortest total travel time. Appendix B provides more details on this procedure.

(iii) Chinese import data

We use transaction-level import data collected by the General Administration of Customs of

[10]This share is calculated using WTO trade in services annual data. See Appendix Table A.1 for a list of destination countries.

Figure 2: Change in Card Transaction Values from 2011 to 2016 in Mainland Chinese Cities



Note: Transaction values are in millions of RMB. The map shows the change in total transaction value between 2011 and 2016 across cities in mainland China.

the People's Republic of China for 2011–2016. Each record includes the product name, HS 8-digit product code, country of origin, date (month and year), transaction value, importing company's name, and city location.[11] We classify products into consumer and non-consumer goods using the United Nations' Broad Economic Categories (BEC).

2.2 Descriptive Statistics

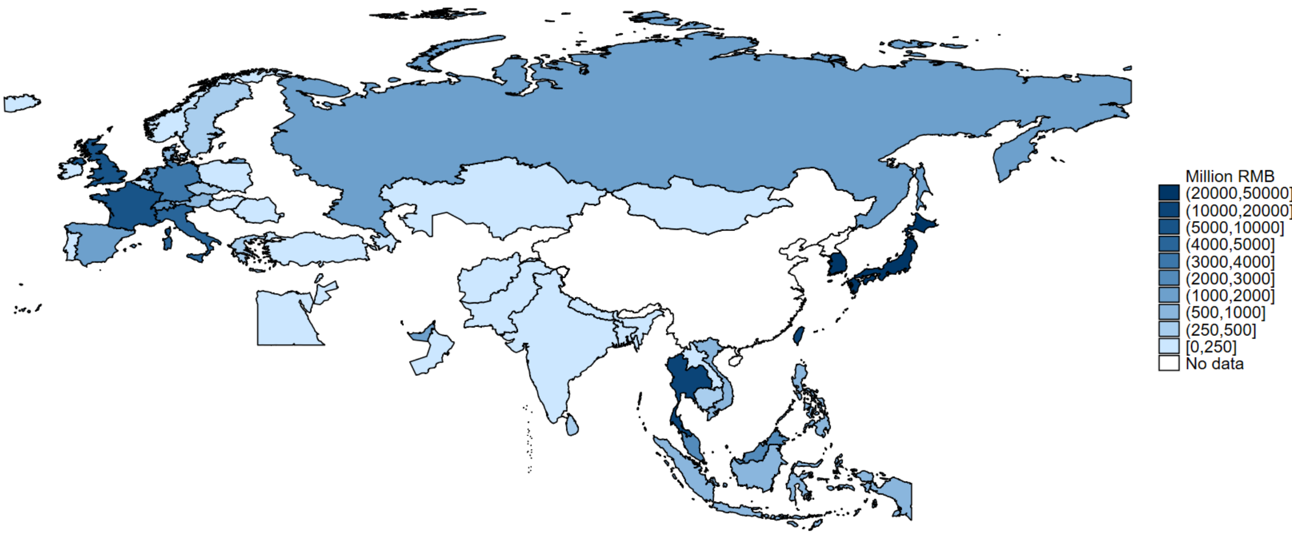
We merge the two main datasets, the Chinese card transaction data and the flight data. There are 9,859 origin-destination pairs in the final dataset, including 193 unique Chinese cities (origins) and 71 unique foreign countries (destinations).[12] Our dataset includes only Chinese cities with airports.

Table 1 reports summary statistics. The average annual transaction value is 14.17 million RMB (approximately 2.31 million USD, based on the 2023 exchange rate), though the distribution is highly skewed due to high-volume travelers from major cities. Similarly, the frequency of direct flights is right-skewed, with an average of 0.61 weekly direct flights and an average travel time of 12.75 hours. The travel time sample is smaller because flight paths with up to three stops are not available for some city–country pairs.

[11] City-level information for 2016 is missing. We supplement firm locations using a concordance table.

[12] We focus on prefecture-level cities in mainland China. Of the 336 cities in the card data, some without airports during the sample period could not be matched to flight data.

Figure 3: Change in Card Transaction Values from 2011 to 2016 in Destination Countries



Notes: Transaction values are in millions of RMB. The map shows the change in total transaction value by country.

2.3 Stylized Facts

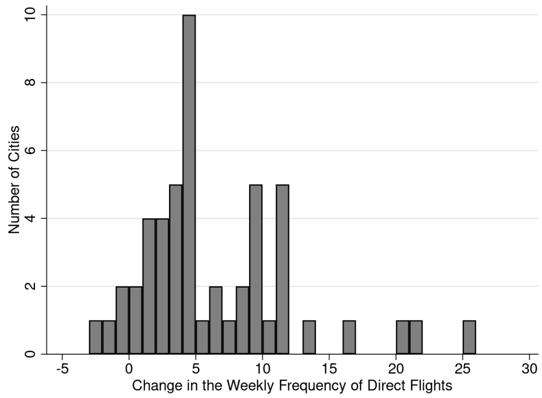
We present three stylized facts that motivate our analysis of how international direct flights affect overseas spending by Chinese travelers.

Fact 1: Regional differences in transaction values

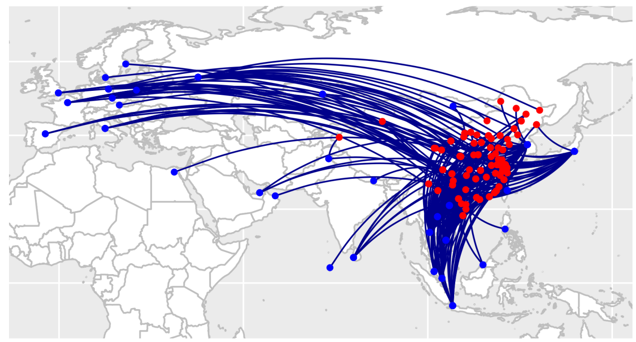
We first analyze the increase in the value of card transactions between 2011 and 2016. Figure 2 shows the change in total transaction value across origin cities in mainland China during this period. All Chinese cities in our sample experienced an increase in transaction value, with particularly large gains in 29 cities where values rose by more than one billion RMB. The cities with the greatest increases were Shanghai (28.2 billion RMB), Beijing (22.5 billion RMB), and Shenzhen (8.0 billion RMB). Notably, substantial growth was observed not only in Eastern coastal cities, but also in inland regions such as Chengdu, Wuhan, and Chongqing. For example, total overseas transactions in Wuhan (an inland city in Hubei Province) increased by approximately 672.8%, from 440.7 million RMB to over 3.4 billion RMB.

We also examine changes in total transaction value across destination countries (Figure 3). All countries in the sample received more card spending in 2016 than in 2011. The largest

Figure 4: Improvements in Air Connectivity Between Chinese Cities and Foreign Countries



(a) Distribution of the Change in Weekly Frequency, 2011-2016



(b) New Direct Flight Routes

Notes: Panel (a) shows the distribution of the number of Chinese cities by the change in the average weekly frequency of direct flights between 2011 and 2016. The sample includes cities with at least one international direct flight in 2011. We calculate the average weekly frequency for each city, then take the difference between 2011 and 2016. In Panel (b), dark blue lines indicate new direct flight routes opened after 2011, connecting Chinese cities (red dots) to destination countries (blue dots).

increases occurred in Japan (41.1 billion RMB), Korea (32.3 billion RMB), and Thailand (17.3 billion RMB). Importantly, growth was not limited to nearby countries; 15 countries experienced increases exceeding one billion RMB, including eight in Europe, such as the UK, France, Italy, and Germany.

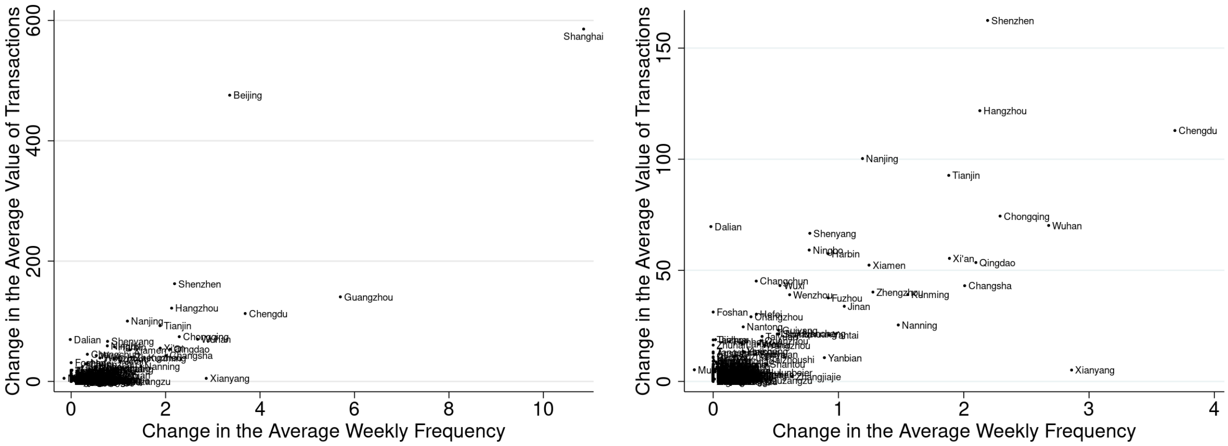
Fact 2: Improvement in air connectivity

Given the rapid growth and regional variation in traveler spending, China presents a useful case study for examining how improvements in air connectivity affect international expenditures. Not only is there considerable demand for outbound travel, but China's aviation network has also expanded dramatically, becoming the world's second-largest air transportation market in 2013 (Gibbons and Wu 2020). We measure air connectivity using the weekly frequency of direct flights (i.e., the number of direct flights per week) at the city-country-year level. Improvements in air connectivity affect spending by Chinese travelers in two ways: (i) more frequent international direct flights on existing routes (intensive-margin effect), and (ii) the addition of new direct routes (extensive-margin effect).

To analyze the intensive-margin effect, we restrict the sample to flight routes that existed in 2011 and compare the average weekly frequency by city between 2011 and 2016.[13] Panel

[13]Flight routes are defined at the city-country level. We take the average frequency within each city.

Figure 5: Change in Air Connectivity and the Value of Transactions



(a) All Chinese Cities

(b) Excluding Three Cities

Notes: Panels (a) and (b) show the correlation between the change in transaction values (y-axis) and the change in the weekly frequency of direct flights (x-axis) from 2011 to 2016. We take the average of transaction values and flight frequencies within each city. Panel (b) excludes Shanghai, Beijing, and Guangzhou.

(a) of Figure 4 shows the distribution of the increase in average frequency in 2016 compared to 2011. Among 51 cities with at least one international route in 2011, airlines operated an additional 6.46 direct flights per week on average by 2016. Air connectivity in Qingdao improved especially through the intensive margin, with an increase of 25.56 weekly flights.[14]

For the extensive-margin effect, Panel (b) of Figure 4 illustrates new direct flight routes introduced between 2011 and 2016. Over this period, 255 new routes were added, and on average each city was connected to 1.32 new destinations. Among 73 cities with at least one new route, Chengdu had the largest increase, establishing direct connections with 12 new countries.[15] Thailand received the highest number of new routes overall, with connections to 39 different Chinese cities.

Fact 3: Positive relationship between air connectivity and card transaction values

Facts 1 and 2 show increases in both overseas transaction values and international flight connections. We now combine the datasets to examine whether improvements in air connectivity are associated with increased overseas spending. Specifically, we calculate the average transaction value and the average weekly frequency of direct flights at the city-year level,

[14] The most frequent routes from Qingdao were to Korea and Japan. Four airlines operated more than two daily flights to Korea in 2016.

[15] The most frequent new route from Chengdu was to Vietnam, with 21.1 weekly flights in 2016.

then take the difference between 2011 and 2016. Figure 5 shows the change in average transaction value against the change in flight frequency. In Panel (a), we observe a positive correlation, which suggests that cities with greater improvements in air connectivity also saw higher overseas spending by Chinese travelers. Major cities such as Shanghai, Beijing, and Guangzhou experienced particularly large increases in both connectivity and spending. The relationship remains positive even when these three cities are excluded, as shown in Panel (b).[16] In the following sections, we explore the causal link underlying this positive relationship.

3 Empirical Strategy

We estimate the causal impact of international air connectivity on Chinese travelers' overseas spending using a gravity regression framework. To address potential bias in the OLS estimates, we construct a novel instrumental variable that exploits exogenous variation in international flight capacity from foreign countries.

3.1 Gravity Equation

Our baseline regression specification is

$$\tilde{X}_{ijt} = \gamma_0 + \gamma_1 \tilde{D}_{ijt} + \delta_{ij} + \eta_{it} + \kappa_{jt} + \epsilon_{ijt}, \tag{1}$$

where $\tilde{X}_{ijt}$ denotes the card transaction value of Chinese travelers from city i in country j, and $\tilde{D}_{ijt}$ is the level of air connectivity (measured as the weekly frequency of direct flights between i and j). We include city-country fixed effects δ_{ij} to account for time-invariant unobserved heterogeneity, such as cultural ties, business linkages, and visa policies. Origin city-year fixed effects η_{it} control for origin-specific time-varying factors like income levels, while destination country-year fixed effects κ_{jt} capture destination-specific time-varying factors, such as tourism attractiveness and the price of travel-related services and goods. This specification is derived from the gravity model described in Appendix C.

We apply the inverse hyperbolic sine (arcsinh) transformation to both card transaction values $\tilde{X}_{ijt}$ and the level of air connectivity $\tilde{D}_{ijt}$ to address the presence of zero values in air connectivity. This transformation approximates the natural logarithm while allowing for

[16]For robustness, we exclude transactions from Shanghai, Beijing, and Guangzhou (Section 4.1). The results remain similar to the full sample analysis.

zero values.[17] The estimated coefficient γ_1 can be interpreted as an elasticity: the percentage change in transaction value associated with a one percentage change in the weekly frequency of direct flights (Bellemare and Wichman 2020).[18] Based on the descriptive analysis in Section 2.3, we expect that improvements in air connectivity lead to a larger number of travelers and, consequently, an increase in overseas spending. We cluster standard errors at the city-country level throughout.

3.2 Instrumental Variable Strategy

Our goal is to estimate the causal effect of air connectivity on Chinese consumers' overseas spending. However, the OLS estimate of γ_1 in equation (1) may suffer from bias, potentially due to measurement error or heterogeneous treatment effects. For example, air connectivity may be measured with error, particularly since our measure does not fully capture the number of passengers on each international flight. In addition, there is a concern about reverse causality: direct flight routes are not randomly assigned, and city-country pairs with pre-existing high travel demand are more likely to receive additional flights.

3.2.1 Instrumental Variable

Our instrumental variable (IV) strategy isolates plausibly exogenous variation in the growth of air connectivity induced by foreign countries' expansions in air transportation capacity, weighted by geographical proximity. This instrument allows us to predict the presence of direct flights between a city-country pair. The IV is defined as:

$$Z_{ijt} = \lambda_{jt} \times \ln dist_{ij}, \tag{2}$$

where $\lambda_{jt} = \frac{flight_{jt}}{\sum_j flight_{jt}}$ is the global share of total international flights departing from country j (excluding flights involving China), and $\ln dist_{ij}$ is the natural logarithm of the geographical distance between Chinese city i and country j.[19] To construct λ_{jt}, we divide the number

[17]The arcsinh transformation is defined as $\tilde{x} = arcsinh(x) = \ln(x + \sqrt{x^2 + 1})$. We use this transformation to compute elasticities while allowing for zero values in the weekly frequency of direct flights. We also run 2SLS regressions (introduced later) without the transformation and confirm that the coefficients remain positive and statistically significant.

[18]Bellemare and Wichman (2020) show that the estimated coefficient can be biased when the dependent or independent variables are small. In our case, although the explanatory variable (weekly flight frequency) is relatively small, their simulation results suggest that the resulting bias is negligible and unlikely to significantly affect our conclusions.

[19]We use flights departing from foreign countries to construct the instrument, while our explanatory variable is based on flights from China to foreign countries.

of outbound flights from country j in year t by the total number of global outbound international flights in that year. We exclude flights arriving in and departing from China from both the numerator and the denominator to ensure that the instrument is exogenous with respect to Chinese city-level travel demand. Multiplying λ_{jt} by the bilateral distance $dist_{ij}$ creates city-country-year variation in the instrumental variable, which reflects how foreign air transportation developments differentially affect Chinese cities depending on distance.

The interaction term in our IV captures the differential impact of air transportation developments in foreign countries between two groups of travelers: those in Chinese cities located closer to the foreign country and those farther away. In general, more distant markets tend to have fewer direct flights (Cristea 2023). Airlines face higher operational costs when serving distant routes, so the effect of improved air transportation infrastructure in destination countries is likely to be attenuated by distance.[20] We expect our instrument to be negatively correlated with the frequency of direct flights: countries with a comparative advantage in air transportation are more likely to be connected by direct flights, whereas more distant city–country pairs tend to have fewer such connections. Note that distance affects the dependent variable (card transaction values) only through air connectivity (our endogenous variable), conditional on the origin-destination FEs.

While our IV differs from a standard shift-share instrument because distance is not a share, the IV is conceptually similar to a shift-share IV developed by Autor et al. (2013). In their study, the authors instrument for the change in US imports from China using other countries' imports from China (i.e., a shock to China's comparative advantage in productivity) and regional employment shares, which allocate the impact of the shock across US regions. In contrast, we instrument for Chinese air connectivity using other countries' air connectivity (i.e., a shock to foreign countries' comparative advantage in air transportation technology) and operational costs of serving the route (measured by the distance between Chinese cities and foreign countries). Here, distance serves to distribute the impact of improvements in foreign air connectivity across Chinese cities.

3.2.2 Identification Assumption

Our key identifying assumption is that the share of flights departing from country j, λ_{jt}, is uncorrelated with demand shocks for travel from Chinese cities to country j in year t. This exclusion restriction holds because air connectivity in destination countries is primarily

[20] In the airline industry, regulations limiting pilot working hours raise the cost of long-haul connections and increase the difficulty of establishing direct flights to distant destinations (Campante and Yanagizawa-Drott 2018).

Table 2: Growth in λ_{jt} and Share of Flights from China, 2011–2016

Country	$\hat{\lambda}$	Share in 2011	Share in 2016
Iraq	148.44	0	0.25
Myanmar	115.11	15.64	10.13
Belarus	53.71	0	0.92
Saudi Arabia	52.45	0.16	0.21
Qatar	42.78	1.41	1.44
Vietnam	40.06	10.42	13.72
Kyrgyzstan	36.76	8.62	5.59
Oman	35.59	0	0.03
Turkey	33.39	0.60	0.47
Taiwan	32.81	23.68	24.65
Pakistan	31.69	1.32	1.22
South Korea	31.09	28.73	31.53
United Arab Emirates	28.00	1.59	1.43
Japan	26.82	21.31	22.85
Laos	20.74	15.84	14.12

Note: This table shows the percentage change in λ_{jt} between 2011 and 2016, $\hat{\lambda}$, along with the share of flights from China to destination countries in 2011 and 2016. All values are percentages. Here, λ_{jt} is the component of our IV representing the global share of international outbound flights departing from country j, excluding those involving China. We report the top 15 countries with the largest increase in λ_{jt}. Countries with no direct flights from China in both years are excluded, as they do not affect our regression results due to city–country fixed effects (i.e., no time variation in weekly frequency of direct flights).

determined by the infrastructure and policy decisions of foreign governments, not by Chinese cities. Table 2 lists the top 15 destinations with the largest growth in the component of our IV, λ_{jt}—the global share of international flights departing from country j, excluding those involving China—between 2011 and 2016. Many of these destinations experienced significant improvements in air transportation, often driven by infrastructure projects such as new airport or terminal openings (Table A.3). Such infrastructure expansions generally reflect government-led investment decisions aimed at expanding air connectivity, rather than responses to demand shocks from specific Chinese cities. In some cases, the drivers were not limited to infrastructure expansion but also included other exogenous political events. For example, in Kyrgyzstan, the closure of the U.S. military base at Manas International Airport freed up capacity for civilian flights. In Iraq, the reconstruction of the Baghdad Airport Road—formerly a dangerous, bomb-ridden route—facilitated greater access to the airport and likely contributed to growth in air traffic. Such events represent supply-side shocks that are plausibly exogenous to Chinese travelers' demand.[21]

Table 2 also reports the share of flights from China to the high λ_{jt} countries in 2011 and 2016, measured as the number of direct flights from China relative to total international flights to the destination. Notably, the share of flights arriving from China did not rise substantially, despite significant expansions in these countries' air transportation networks. For example, while the global share of flights departing Iraq and Myanmar increased by over 100%, the share of flights from China to Iraq was zero in 2011 and rose to only 0.25% by 2016. In the case of Myanmar, the share actually declined from 15.64% in 2011 to 10.13% in 2016. These patterns suggest that the objective of air transportation development was to attract travelers globally rather than specifically target Chinese travelers. The variation in international air connectivity is plausibly exogenous to city-level demand shocks in China and serves as a valid instrument for our analysis.

3.3 2SLS Specification

Using our IV defined in equation (2), we estimate the following two-stage least squares (2SLS) system:

$$\tilde{D}_{ijt} = \alpha_0 + \alpha_1 Z_{ijt} + \delta_{ij} + \eta_{it} + \kappa_{jt} + \xi_{ijt} \qquad \text{(first stage)} \tag{3}$$

$$\tilde{X}_{ijt} = \gamma_0 + \gamma_1 \tilde{D}_{ijt} + \delta_{ij} + \eta_{it} + \kappa_{jt} + \epsilon_{ijt}, \qquad \text{(second stage).} \tag{4}$$

The first-stage coefficient, α_1, captures the relationship between the IV, Z_{ijt}—composed of the global share of flights departing from foreign country j and the distance between country j and Chinese city i—and the degree of air connectivity between the two locations, $\tilde{D}_{ijt}$. The exclusion restriction discussed above holds if the IV, Z_{ijt}, is uncorrelated with other unobserved determinants of card transaction values, ϵ_{ijt}. The second-stage coefficient of interest, γ_1, then identifies the causal effect of air connectivity on card transactions made by travelers from city i in destination country j.

[21] For further robustness, we show that growth in international outbound flights from destination countries is better explained by supply-side factors than by demand-side factors. See Appendix Section A.4.

4 Results

We estimate the impact of air connectivity on traveler spending using our IV framework. First, we report 2SLS results that identify the causal effect of the weekly frequency of direct flights on card transaction values. We then examine the robustness of our estimates across different specifications and use travel time as an alternative measure of air connectivity. In the last subsection, we use the estimated coefficient to forecast the potential impact of a new airport opening on traveler spending.

4.1 Main Result

Table 3 presents the results from our main specification. Column 2 reports the IV regression estimates, which include three types of fixed effects (FEs): origin city-year FEs, destination country-year FEs, and city-country pair FEs. The IV coefficient on the frequency of direct flights is positive and significant. Specifically, a 1% increase in the weekly frequency of direct flights leads to a 1.2% increase in cross-border travel spending. The average weekly frequency of direct flights is 0.61 over the data period. Intuitively, our regression implies that in a city with the average weekly frequency, adding one direct flight per month raises traveler spending in the destination country by 49.18%. First-stage results are reported at the bottom of the table. The coefficient on the IV is negative and highly significant. Importantly, the first-stage F statistic is 16.98, which indicates that we can reject the null hypothesis of a weak instrument.

The IV coefficient is larger than the OLS coefficient reported in column 1. This downward bias does not rule out reverse causality, but it suggests that other forces may be weakening the estimated relationship between air connectivity and the value of card transactions. One such force is attenuation bias due to measurement error in air connectivity. Another possible explanation is treatment effect heterogeneity. The difference between our 2SLS and OLS coefficients highlights that the two estimators identify different parameters. The OLS estimate reflects an average effect across all city-country routes, including those where existing air connectivity is already dense and additional flight frequency yields diminishing returns on travel demand. In contrast, our 2SLS estimate captures the local average treatment effect (LATE)—the causal impact of improved air connectivity for city-country pairs whose connectivity changes in response to exogenous developments in air transportation capacity near their location. These destinations are often relatively underserved in terms of direct air access to China, and therefore exhibit larger marginal responses to improved connectivity. A

Table 3: Main Results

	Transaction value		
	Baseline sample		Excl. big cities
	OLS	2SLS	2SLS
	(1)	(2)	(3)
Weekly frequency	0.019	1.204***	1.232***
	(0.022)	(0.413)	(0.394)
Origin city-year FEs	Yes	Yes	Yes
Destination country-year FEs	Yes	Yes	Yes
City-country FEs	Yes	Yes	Yes
Observations	42,940	42,940	41,887
First stage			
IV		−39.568***	−42.165***
		(9.602)	(9.627)
F-statistic		16.981	19.184

Notes: Columns 1 and 2 report the baseline results. The specification in column 3 excludes city–country pairs that include Shanghai, Beijing, or Guangzhou. Standard errors, clustered at the city-country level, are in parentheses. Significance levels are $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

new flight likely causes a spike in demand, which is particularly relevant in many policy settings, whereas existing (and possibly saturated) routes captured in OLS contribute smaller marginal effects and thus lower the OLS estimate.

In column 3, we restrict the sample to test whether our results are sensitive to sample selection. One potential concern is that a large share of Chinese international travelers originate from three major cities—Shanghai, Beijing, and Guangzhou—as shown in Figure 5.[22] To test this possibility, we drop all city-country pairs involving these three cities and re-estimate our 2SLS specification. Column 3 of Table 3 shows that the resulting 2SLS coefficient is very similar to the one from the full sample (column 2) in magnitude, statistical significance, and sign. This finding suggests that our main result is not driven by travelers from the three largest Chinese cities.

[22]Table A.2 shows that top city–country pairs involving Shanghai and Beijing account for only a small share of total transactions, so omitting them does not affect our analysis. Our results are not driven by the extensive travel between major cities like Beijing or Shanghai and nearby countries such as Japan and South Korea.

Table 4: Robustness Checks using Frequency of Direct Flights

	Transaction value		
	European regions 2SLS (1)	Cities w/o airports 2SLS (2)	Nearby airports 2SLS (3)
Weekly frequency	1.177***	2.643***	1.636***
	(0.394)	(0.900)	(0.537)
Origin city-year FEs	Yes	Yes	Yes
Destination country-year FEs	Yes	Yes	Yes
City-country FEs	Yes	Yes	Yes
Observations	42,940	71,323	71,323
First stage			
IV	−40.474***	−22.905***	−37.003***
	(9.534)	(6.829)	(11.694)
F-statistic	18.021	11.249	10.012

Notes: In column 1, we group European countries into four regions: Central and Eastern Europe, Northern Europe, Southern Europe, and Western Europe. In column 2, we include cities without airports in our baseline sample. In column 3, we consider all airports located within 200 km of each Chinese city. Standard errors, clustered at the city-country level, are in parentheses. Significance levels are $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 Robustness Checks

We examine the robustness of our results to alternative sample constructions and connectivity measures. Specifically, we test whether our main findings hold under different definitions of the sample and when using an alternative measure of air connectivity based on travel time rather than flight frequency.

4.2.1 Alternative Specifications with Weekly Frequency of Direct Flights

As a robustness check, we use alternative specifications based on different sample constructions. First, we consider the possibility that Chinese travelers visit other countries close to their initial arrival country, a pattern especially relevant in Europe, where multiple destinations are easily accessible. To address this concern, we group European countries into four regions and measure the frequency of direct flights at the regional level.[23] Column 1 of Table 4 shows the result from this specification, and the estimated coefficient closely aligns with our main regression result.

Next, we expand the scope of Chinese cities in the sample. Our main specification focuses on cities with airports during our sample period. In columns 2 and 3, we add an additional group of Chinese cities—those without airports. Including these cities is expected to increase the size of the coefficient, as the baseline group now consists of cities without access to any air transportation, rather than cities without access to international flights. Column 2 of Table 4 shows that the 2SLS estimate is positive and significant, and, as expected, larger than the main result reported in column 2 of Table 3.

We also consider a scenario where Chinese travelers have the option to use airports in neighboring cities. Specifically, we assume that travelers are willing to travel up to 200 km to use another airport, and we include all flights departing from airports within a 200 km radius of each city center.[24] In our main regressions, international direct flights are assumed to influence only Chinese travelers in cities with an airport. In this alternative specification, each flight route serves a broader population of potential travelers, as nearby cities are also assumed to benefit. Because each international flight route now affects more cities, the associated change in spending is larger. The regression result supports this intuition: we observe a larger coefficient on the weekly frequency of direct flights (column 3 of Table 4).

4.2.2 Travel Time

We consider travel time from Chinese cities to foreign countries as an alternative measure of air connectivity. While our main specification relies on the frequency of international direct flights, this alternative measure incorporates both direct and indirect connections. Travel time is measured in hours. The IV regression results are presented in column 2 of Table 5. The coefficient of travel time is -2.02 and significant at the 1% level, which indicates that a 1% decrease in travel time leads to a 2.02% increase in transaction value. In our data, a new direct flight route reduces travel time by 1.87 hours on average.[25] Given an average travel time of 12.75 hours in our data, this result suggests that if travelers in a city with the average travel time gain access to a new direct flight route (i.e., saving 1.87 hours), the

[23] According to the European Union, the regions are defined as follows: Central and Eastern Europe, Northern Europe, Southern Europe, and Western Europe.

[24] Nearby airports are identified based on city coordinates (longitude and latitude).

[25] On average, a new flight route reduces travel time by 3.31 hours by providing a non-stop flight to a destination (i.e., direct effect), while it enables a traveler to save 0.86 hours (52 minutes) using the new non-stop destination as a transit airport (i.e., indirect effect).

Table 5: Regressions with Travel Time

	Transaction value		
	All cities		Excl. big cities
	OLS	2SLS	2SLS
	(1)	(2)	(3)
Travel time	-0.164***	-2.016**	-2.215**
	(0.028)	(0.967)	(0.993)
Origin city-year FEs	Yes	Yes	Yes
Destination country-year FEs	Yes	Yes	Yes
City-country FEs	Yes	Yes	Yes
Observations	34,584	34,584	33,531
First stage			
IV		17.532***	18.064***
		(6.236)	(6.333)
F-statistic		7.904	8.135

Notes: This table reports regression results using travel time, rather than the weekly frequency of direct flights, as the measure of air connectivity. Columns 1 and 2 present the main results using all cities. Column 3 excludes city–country pairs involving Shanghai, Beijing, or Guangzhou. Standard errors, clustered at the city-country level, are in parentheses. Significance levels are *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

transaction value will increase by 29.57%. A similar and statistically significant estimate is obtained when excluding major cities such as Beijing, Shanghai, and Guangzhou (column 3 of Table 5).

This alternative specification helps capture the broader impact of new direct flights, which can reduce travel time not only to the directly connected destination but also to third countries via improved transfer routes. For example, there was no direct flight from Chongqing to the UK between 2011 and 2015. In 2011, the shortest route involved flying from Chongqing to Beijing and then connecting to the UK. In 2012, a new direct route from Chongqing to Finland was introduced, which enabled travelers to reach the UK via Finland and reduced travel time to the UK by an hour. Finally, in 2016, another new direct flight from Chongqing to the UK became available, which allowed travelers to visit the UK four hours faster without any connections.

The first-stage F statistic for the specification using travel time is lower than that for the main specification using the frequency of direct flights. This difference reflects the nature of our IV: it captures variation driven by expansions in air connectivity in destination countries, which increases the likelihood of new or more frequent direct connections to those countries. While this instrument strongly predicts direct flight frequency, it is less predictive of travel time because new routes may reduce travel time not only to the destination itself but also to third countries via improved transfer options. As a result, the instrument has weaker explanatory power for travel time.

4.3 Policy Implications

Our main result indicates that a 1% increase in the weekly frequency of direct flights increases the value of card transactions by 1.2%. This information helps policymakers evaluate the potential impact of improving air transportation facilities on inbound visitor spending. In this section, we evaluate the economic implications of a major airport investment by focusing on the opening of Istanbul Airport in Turkey and comparing the cost of the investment with the estimated benefits from increased visitor spending. Istanbul Airport opened in October 2018 and became fully operational in April 2019. It is now the busiest airport in Europe with six runways and a capacity to accommodate 500 aircraft.

This expansion in air transportation capacity led to three new flight schedules between Chinese cities and Istanbul. One of the new routes connected Beijing and Istanbul with three weekly flights, operated by China Southern Airlines beginning in December 2018.[26] Before the new airport opened, Turkish Airlines operated direct flights between Beijing and Istanbul every day. Therefore, the new three-weekly flight schedule increased the weekly frequency of direct flights from 7 to 10.

We forecast the impact of the additional three-weekly flights from Beijing on traveler spending in Turkey. We obtain the projected transaction value for 2019, X_{2019}, using the equation:

$$X_{2019} = X_{2016} \times (1 + 1.2\Delta_{frequency}),$$

where X_{2016} is the value of card transactions in 2016 (observed in our dataset), $\Delta_{frequency}$ is the change in the weekly frequency of direct flights, and 1.2 is the coefficient in column 2 of Table 3. Because the frequency increased from 7 to 10, $\Delta_{frequency} = (10 - 7)/7 = 0.43$. We estimate that the transaction value is projected to increase by 3.16 million USD after the new airport opening in 2019 (i.e., $X_{2019} - X_{2016} = 3.16$).

How much did Turkey invest to attract additional direct flights from Beijing? The to-

[26] China Southern Airlines also launched a new route between Wuhan and Istanbul in May 2019. Additionally, China's Sichuan Airlines began a new flight schedule connecting Chengdu to Istanbul in April 2019.

tal investment in Istanbul Airport amounted to 12 billion USD. The additional flights from Beijing to Istanbul account for 2% of the total new flights from all overseas airports to Istanbul in 2019. Although the long-term economic benefits are challenging to estimate, we refer to the cost and benefit analysis for the Heathrow Airport expansion made by the UK government and consider that the new airport will contribute to the economy for 60 years.[27] The estimated cost for Turkey to attract additional travelers from Beijing in 2019 is 4 million USD (12 billion USD/60 years $\times$ 0.02), which is comparable in magnitude to the estimated benefits from traveler spending on this route (3.16 million USD as mentioned above). The new airport construction brings additional benefits beyond visitor spending, including job creation, enhanced business relationships, increased cargo capacity, and improved regional accessibility. Overall, our analysis suggests that traveler spending alone can offset a meaningful portion of infrastructure costs, and the full economic impact of air connectivity improvements is likely to be substantial.

5 Further Analysis

We show that the developments in air transportation networks increase inbound visitor spending using cross-border card transaction data. In this section, we further investigate our findings on card transactions by comparing them to the effect on goods trade. Additionally, we study how shocks to consumer preferences toward destination countries affect traveler spending and trade in goods.

5.1 Trade in Goods

Improvements in air transportation networks affect not only consumer travel but also trade in goods through two main channels: a direct effect via increased freight capacity and an indirect effect via travelers' exposure to foreign products. Approximately half of air freight is shipped on passenger aircraft alongside passengers and baggage, although individual shipments are small relative to those on cargo flights.[28] Additionally, an increase in outbound travelers may generate positive spillovers for goods imports—particularly consumer goods—as travelers discover new products abroad and continue purchasing them after returning

[27]Reference: The Government of the United Kingdom, Department for Transport Heathrow Northwest Runway, Economic Benefits (link, accessed on August 18, 2025)

[28]For example, a Boeing 747-400, one of the largest passenger planes, can transport 5,330 cubic feet of cargo (the same amount as two semi-truck trailers) together with 416 passengers (link, accessed on January 21, 2023). Also, see "Inbound air freight prices go sky high in the midst of pandemic," the US Bureau of Labor Statistics (link, accessed on January 21, 2023).

Table 6: Effect of Air Connectivity on Imports

	Value of imports		
	Total	Consumer goods	Non-consumer goods
	(1)	(2)	(3)
Weekly frequency	−0.269	1.486***	−0.839
	(0.525)	(0.483)	(0.540)
Origin city-year FEs	Yes	Yes	Yes
Destination country-year FEs	Yes	Yes	Yes
City-country FEs	Yes	Yes	Yes
Observations	42,940	42,940	42,940

Notes: This table presents the results of IV regressions using the value of imports as the dependent variable. The first-stage regression results are the same as those reported in column 2 of Table 3. Standard errors, clustered at the city-country level, are shown in parentheses. Significance levels are $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

home. An increase in the weekly frequency of direct flights can therefore affect trade in goods both directly and indirectly.

Building on this idea, we extend our analysis to examine the effect of air connectivity on imports from country j to Chinese city i by running IV regressions using customs data instead of card transactions. We classify goods into consumer and non-consumer categories, with consumer goods accounting for about 6% of total imports. Table 6 reports the results for total imports (column 1), consumer goods imports (column 2), and non-consumer goods imports (column 3). The coefficient on the weekly frequency of direct flights is not statistically significant for total imports. However, the coefficient is positive and statistically significant for consumer goods, while insignificant for non-consumer goods. This pattern suggests that the impact of air connectivity depends on the type of goods: connectivity enhances imports of consumer goods—final products directly consumed by households, whose preferences are more sensitive to accessibility—but not imports of non-consumer goods. In particular, improvements in air connectivity appear to have stronger effects on certain consumer goods, such as food, pharmaceuticals, and optical goods (Appendix A.5).

5.2 Effects of Political Conflicts

Unlike goods trade, the cost of travel depends not only on trade costs (measured here by air connectivity) but also on the amenities provided by each destination country. We examine how these two factors—trade costs and destination attractiveness—interact to shape international traveler spending. To identify exogenous shocks to consumer preferences, we exploit political conflicts between destination countries and China. We expect that hostile sentiment toward a particular destination weakens the effect of air connectivity on cross-border travel, while having little influence on goods trade, especially non-consumer goods.

During our data period, there are four notable conflicts between China and Japan, the Philippines, South Korea, and Norway.[29] First, a political conflict arose in 2012 between China and Japan over the Diaoyu Islands (also known as the Senkaku Islands), which led to a series of anti-Japanese demonstrations, including consumer boycotts of Japanese products across many Chinese cities. Second, tensions between China and the Philippines escalated in 2012 over Huangyan Island (Scarborough Shoal). In response, China issued a document aimed at strengthening the inspection and quarantine of fruit imports from the Philippines. Third, in 2016, the South Korean and US governments announced their agreement to deploy the Terminal High-Altitude Area Defense (THAAD) system on the Korean Peninsula.[30] China opposed the deployment and imposed sanctions on travel and trade with South Korea. Fourth, the Norwegian Nobel Committee awarded the Nobel Peace Prize to the Chinese human rights activist Liu Xiaobo in late 2010. The Chinese government strongly denounced the award and introduced political and economic sanctions against Norway, which began in mid-December 2010 and had their main impact in 2011.

We study the effect of these four political conflicts on Chinese traveler spending and goods imports. First, we construct an indicator of boycotts, $\mathcal{I}[Boycott_{jt}]$, which takes the value of one for a country involved in a conflict in the year of the event. Specifically, the indicator equals one for Japan in 2012, one for the Philippines in 2012, one for South Korea in 2016, and one for Norway in 2011.[31] We then introduce the interaction term between the weekly frequency of direct flights and $\mathcal{I}[Boycott_{jt}]$ into the 2SLS specification.[32]

The results are presented in Table 7. We find a negative and statistically significant coefficient on the interaction term between air connectivity and boycotts (column 1). This result indicates that during a political conflict between China and a destination country,

[29]Recent studies show an adverse effect of political conflict on trade between China and Japan (Heilmann 2016), the Philippines (Luo et al. 2021), South Korea (Kim and Lee 2021), and Norway (Kolstad 2020).

[30]THAAD is a missile defense system designed to intercept and destroy ballistic missiles, and is intended as a countermeasure against North Korea's nuclear and missile threats.

[31]There are no direct flight schedules from Chinese cities to Norway during our data period. Travelers from China to Nordic countries often begin their regional trips from Finland (link, accessed on January 23, 2023). We use flights to Finland instead as a proxy.

[32]We do not include the indicator variable $\mathcal{I}[Boycott_{jt}]$ in the regression, as it is absorbed by the country-year FEs.

Table 7: Air Connectivity and Boycotts

	Card transactions	Goods imports		
		Total	Consumer	Non-consumer
	(1)	(2)	(3)	(4)
Weekly frequency	1.361***	−0.288	1.560***	−0.896
	(0.462)	(0.566)	(0.532)	(0.586)
Weekly frequency × Boycotts	−0.177***	0.021	−0.086*	0.066
	(0.045)	(0.050)	(0.050)	(0.053)
Origin city-year FEs	Yes	Yes	Yes	Yes
Destination country-year FEs	Yes	Yes	Yes	Yes
City-country FEs	Yes	Yes	Yes	Yes
Observations	42,940	42,940	42,940	42,940

Notes: This table presents the results of IV regressions examining the impact of political conflicts on the effect of air connectivity. The dependent variable is card transactions in column 1, total goods imports in column 2, consumer goods imports in column 3, and non-consumer goods imports in column 4. The first-stage F statistic is 14.98. Standard errors, clustered at the city-country level, are in parentheses. Significance levels are $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

the positive effect of air connectivity on traveler spending decreases by 0.18 percentage points compared to periods without such conflicts. For goods imports, the coefficient on the interaction term is not statistically significant in the regressions with total imports or non-consumer goods (columns 2 and 4). In contrast, the coefficient is negative and statistically significant for consumer goods (column 3), consistent with the idea that consumer goods are more closely tied to consumer preferences than non-consumer goods. Overall, our findings suggest that political conflicts—negative shocks to consumer preferences—reduce the positive effect of air connectivity on cross-border travel but not on goods trade, particularly non-consumer goods.

6 Conclusion

This study investigates the impact of air connectivity on international traveler spending using unique data on card transactions from Chinese cities to foreign countries. We develop a novel instrument for air connectivity based on destination countries' comparative advantage in air transportation. Our results indicate that a 1% increase in the weekly frequency of direct flights between a Chinese city and a foreign country leads to a 1.2% increase in the value of transactions in the destination country.

We also evaluate a recent investment in air transportation infrastructure by using the opening of Istanbul Airport as an illustrative case. The analysis suggests that inbound visitor spending makes a meaningful contribution to offsetting the cost of investment, even without accounting for broader economic benefits. Furthermore, we extend our analysis to examine the effect of air connectivity on trade in goods and how shocks to consumer preferences (such as boycotts) differentially affect traveler spending and goods trade. We find that boycotts weaken the positive effect of air connectivity on cross-border travel, but not on trade in goods, particularly non-consumer goods.

Our findings provide new insights into the relationship between investment in air connectivity and traveler spending, with implications for policies aimed at promoting countries as attractive destinations. To fully assess the benefits of such investments, policymakers should account for the mediating role of local consumer preferences, including cultural ties and attitudes toward destination countries. Our results suggest that fostering cultural exchanges and cultivating welcoming attitudes toward inbound travelers can amplify the gains from improved air connectivity.

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Appendix A Appendix Tables

A.1 Destination Countries

There are 71 unique foreign countries in our final dataset. The travel destinations in the data are mainly the countries of the Belt and Road Initiative in Eurasia, along with Japan and EU countries.

Table A.1: Destination Countries

Afghanistan	Austria	Azerbaijan	Bahrain
Bangladesh	Belarus	Belgium	Brunei
Bulgaria	Cambodia	Czech Rep	Denmark
Egypt	Estonia	Finland	France
Georgia	Germany	Greece	Hungary
Iceland	India	Indonesia	Iraq
Ireland	Israel	Italy	Japan
Jordan	Kazakhstan	Kuwait	Kyrgyzstan
Laos	Latvia	Lebanon	Luxembourg
Malaysia	Maldives	Malta	Monaco
Mongolia	Myanmar	Nepal	Netherlands
Norway	Oman	Pakistan	Philippines
Poland	Portugal	Qatar	Romania
Russian Federation	Saudi Arabia	Singapore	Slovakia
Slovenia	South Korea	Spain	Sri Lanka
Sweden	Switzerland	Tajikistan	Taiwan
Thailand	Timor-Leste	Turkey	United Arab Emirates
United Kingdom	Uzbekistan	Vietnam	

Note: The table lists the travel destinations in our data. See Section 2.1 for details.

A.2 City-Country Pairs With the Largest Transaction Values

The three biggest Chinese cities—Beijing, Shanghai, and Guangzhou—have the highest numbers of direct flights and the greatest value of card transactions (Figure 5). A potential concern is that transaction values might be heavily concentrated in only a few city–country pairs. Table A.2 lists the five Chinese city–foreign country pairs with the largest average transaction values. The largest flows originate from the major Chinese cities (Shanghai and Beijing) to major nearby economies such as Japan, South Korea, and Taiwan. However, these city–country pairs account for only a small share of the total. For example, transactions from Shanghai to Japan represent just 4.9% of the total on average. This suggests that the transaction values are not overly concentrated in a few city-country pairs.

Table A.2: City-Country Pairs With the Five Largest Transactions

City	Country	Average (yearly)	Share
Value of transactions (in million RMB):			
Shanghai	Japan	4,969.72	0.049
Shanghai	South Korea	3,113.90	0.031
Beijing	Japan	3,087.32	0.030
Beijing	South Korea	2,925.83	0.029
Shanghai	Taiwan	2,264.65	0.022

Note: This table shows five Chinese city–foreign country pairs with the largest average transaction values. The average is calculated over the full sample period for each pair. Shares represent the average value of each city–country pair relative to the total average transaction value, which is 101,596.3 million RMB.

A.3 Key Events in Air Transportation Development

One concern regarding our IV is whether the development of air transportation—measured as the increase in the total number of outbound flights—is driven by demand or supply. Table A.3 lists major events during 2011–2016 that are likely to have contributed to increases in the total number of flights departing from these countries. All events represent supply-side shocks, such as infrastructure expansions, policy initiatives, or historical circumstances, that are plausibly exogenous to Chinese travelers' demand.

Table A.3: Key Events in Air Transportation Development, 2011–2016

Country	Event
Iraq	Reconstruction of the Baghdad Airport Road, once a dangerous, bomb-ridden route
Myanmar	Expansion of Yangon International Airport
Belarus	Minsk National Airport's membership in the International Airports Association
Saudi Arabia	Government program Saudi Vision 2030, prioritizing tourism growth and aviation infrastructure modernization
Qatar	Opening of Hamad International Airport
Vietnam	New international terminal openings in Hanoi and Da Nang
Kyrgyzstan	End of US military use of Manas International Airport; launch of new carrier Air Manas
Oman	Opening of a new runway at Muscat International Airport
Turkey	Government's Vision 2023 aviation expansion plan, including the opening of new regional airports
Taiwan	Airport expansions at Taoyuan; Tainan and Chiayi airports gained international airport status
Pakistan	Expansion at Islamabad's Benazir Bhutto International Airport; new terminal opening at Multan International Airport
South Korea	Expansions at Incheon and Gimpo airports
United Arab Emirates	Expansion of Dubai International Airport; opening of Al Maktoum International Airport
Japan	Government-led resumption and expansion of international services at Haneda Airport
Laos	Expansions at Wattay International Airport and Luang Prabang International Airport

Note: This table lists major country-level events during 2011-2016 that are likely to have contributed to the development of air transportation, measured by the number of flights departing from country j. The countries shown are the top 15 with the largest increase in λ_{jt}, as reported in Table 2.

A.4 Drivers of International Flight Growth

One potential concern with our identification strategy is whether developments in air transportation capacity in destination countries are driven by supply-side or demand-side shocks. Table A.3 lists the countries with substantial growth in air transportation capacity, where this growth coincided with new airport or terminal openings, policy changes, or historical events. We conduct an additional robustness check to assess whether growth in international outbound flights from destination countries is better explained by supply-side factors rather than demand-side factors.

Increases in flight volumes can result from either supply or demand changes. A supply-driven increase reflects an actual expansion in capacity (e.g., airport or terminal expansions), whereas a demand-driven increase stems from rising air travel demand, which can lead to congestion and higher gate fees. In congested airports, operators may also shift toward attracting more long-haul flights over short-haul flights due to higher landing fees. To test this possibility, we examine whether congestion levels in destination countries are correlated with the ratio of long-haul to short-haul flights. Specifically, we estimate:

$$\text{Long-haul flight ratio}_{jt} = \theta_0 + \theta_1 \log(\text{total outbound flights}_{jt}) + \alpha_t + \alpha_j + \epsilon_{jt},$$

where Long-haul flight ratio$_{jt}$ is the ratio of the number of long-haul flights to short-haul flights in country j, and total outbound flights$_{jt}$ is the total number of international flights departing from country j in year t (the numerator of λ_{jt}). α_t and α_j are year and destination country fixed effects.[33] While total outbound flights do not directly measure airport congestion, destination country fixed effects control for time-invariant capacity differences across countries. Our identification relies on within-country changes over time, which are plausibly correlated with congestion levels if capacity constraints bind. The analysis covers our sample period, 2011–2016.

The regression results are reported in Table A.4. The estimated coefficient θ_1 is statistically insignificant across all specifications, suggesting that airports in destination countries do not systematically alter the mix of short-haul and long-haul flights in response to changes in the total number of flights. This finding supports the view that the observed increases in total outbound flights (the numerator of λ_{jt}) are unlikely to be driven by congestion at destination airports—a demand-side factor—and are instead more consistent with supply-side shocks, as discussed in the main text.

[33] Following Campante and Yanagizawa-Drott (2018), we define long-haul flights as those covering at least 9,700 kilometers (approximately 6,000 miles).

Table A.4: Outbound Flight Volume and Composition of Long- vs. Short-haul Flights

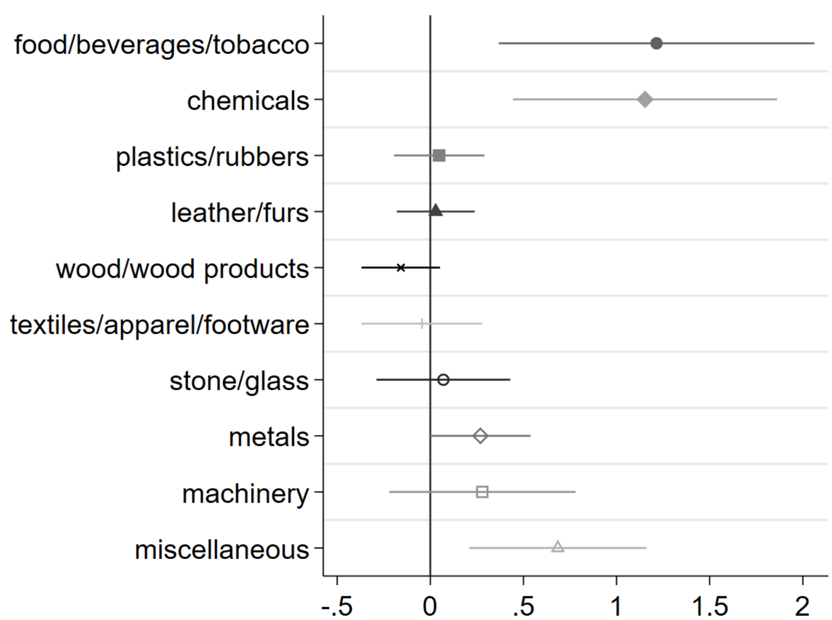
	Long-haul flight ratio$_{jt}$		
$\log(flight_{jt})$	0.002	0.002	0.003
	(0.001)	(0.001)	(0.005)
Year FEs	No	Yes	Yes
Destination country FEs	No	No	Yes
R-square	0.031	0.041	0.914
N	372	372	369

Notes: Standard errors, clustered at the destination country level, are in parentheses. Significance levels are $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.5 Heterogeneity across Different Types of Consumer Goods

We examine the effect of air connectivity on consumer goods imports across different industries, as shown in Figure A.1. Specifically, we find that imports of food, beverages, and tobacco, as well as chemical products (primarily pharmaceuticals and cosmetics), increase by more than 1% in response to a 1% increase in the weekly frequency of direct flights. The effect of air connectivity on miscellaneous goods—including photographic apparatus, optical goods, and watches—is also positive and statistically significant. These products are predominantly transported by air. For example, food and beverages are perishable and time-sensitive (Djankov et al. 2010), while pharmaceuticals, as well as photographic apparatus, optical goods, and watches, have high unit values and are primarily shipped via fast but costly air transport.[34]

Figure A.1: Estimated Coefficients on Consumer Goods Imports by Industry



Notes: This figure shows the coefficients estimated from regressions of consumer goods imports on air connectivity. Regressions are run separately for different sub-industry categories over the period 2011–2016.

[34]According to Harrigan (2010), 65% of medical and pharmaceutical products and 74% of photographic apparatus, optical goods, and watches were imported by air into the US in 2003.

Appendix B Data Appendix

B.1 Card Transaction Data

- Chinese cities
  - Chinese cities in our data are four centrally administered municipalities (Beijing, Tianjin, Chongqing, and Shanghai) and the prefecture-level cities, including 292 prefecture-level cities, seven regions, 30 autonomous prefectures, and three leagues (see the Administrative Divisions of the People's Republic of China, link for the details). Two prefecture-level cities, Sansha and Danzhou, are not included in our transaction data. In total, our dataset comprises 336 Chinese cities, including 193 with airports.
- Data in 2016
  - We address the missing data from September to December 2016 using the monthly transaction data in 2015. This approach allows us to account for the seasonality of travel demand across different regions. The monthly transaction data are aggregated at the China-foreign country level (unlike the yearly data, the monthly data are not disaggregated by Chinese city).
  - Because few monthly transactions are recorded in some foreign countries, we aggregate countries into sub-regions based on the geographic regions defined by the United Nations (link). For each sub-region, we calculate the share of transaction values made from January to August relative to the total value. On average, the share is 66%. A higher proportion of travelers visit Western Europe between January and August (around 70% of the total transactions in 2015) compared to Southern Asia and Eastern Europe. Specifically, we calculate for each sub-region:

$$Expected[x_{\{t=Jan-Dec,\ 2016\}}] = x_{\{t=Jan-Aug,\ 2016\}} \times \frac{x_{\{t=Jan-Dec,\ 2015\}}}{x_{\{t=Jan-Aug,\ 2015\}}},$$

    where x_t is the value of transactions over the period, t.
  - Egypt is the only Northern African country in our data, so we added it to the Western Asian region (which includes Saudi Arabia).

B.2 Global Flight Schedules

- OAG Analyzer
  - The data is at the air-carrier level. We restrict the data to passenger flights (excluding cargo flights) and flights operated by operating carriers (excluding code-share flights).
  - We consider all flights arriving in destination countries, regardless of the cities of arriving airports.
  - We add names of the cities with airports to the original data using the table provided by OAG. We also searched on the internet to identify airports serving two cities. The following airports serve two cities: Yangzhou Taizhou International Airport (serving Taizhou and Yangzhou), Xining Caojiabao International Airport (serving Haidong and Xining), Xi'an Xianyang International Airport (serving Xianyang and Xi'an), and Lhasa Kongga International Airport (serving Shannan and Lhasa).
  - OAG records flights with stopovers, not just direct flights. For flights with stopovers at domestic airports before departing for foreign countries, we keep only the flights bound for foreign countries. For example, a flight route from Chongqing to Japan connects through Shanghai. We focus on direct flights connecting a Chinese city with a foreign country (e.g., flights from Shanghai to Japan) and exclude domestic stopover flights (e.g., flights from Chongqing to Japan).
  - Regarding flights with stopovers at international airports, we use only the flights directly connecting Chinese cities with foreign countries. For example, there is a stopover flight departing from Guangzhou to Sri Lanka via Thailand. We keep only the flight from Guangzhou to Thailand.
- Weekly frequency of direct flights
  - We calculate the sum of frequencies in each Chinese city-destination country-year and divide the total by 52 to measure the weekly frequency.
- Travel time
  - We identify the most frequent flight routes to destination countries. For example, flights from Beijing to Japan arrive at different airports such as Narita, Haneda, Osaka, New Chitose (located in Hokkaido), and Naha (located in Okinawa). We

use the travel time of the flight between Beijing and Narita because flights to Narita are the most frequent from Beijing.

– There are some extremely short elapsed times in the data. We exclude the route if the data suggest that a plane travels at speeds exceeding 1,000 km/hour.

– Even on the same flight route, the elapsed time can vary slightly in different flight schedules. We calculate the average flight time using the frequency of flights as the weight. This allows us to have the same elapsed time on the same flight routes over the data period.

– After cleaning the data, we consider all possible flight routes with up to three stops (two stops at domestic airports and one stop at a foreign airport). We add a three-hour layover time for every stop to the total elapsed time. We then select the route with the shortest travel time among all of the possible flight routes.

– We keep city-country pairs with travel time recorded over our data period.

Appendix C Model

Our model is based on Head et al. (2009), who introduce a gravity-type equation for bilateral service trade. Each foreign country offers amenities for travelers, and a consumer makes a discrete choice among her possible destinations based on her preferences. We refer to Faber and Gaubert (2019) to set up consumers' utility for international travel.

Consumer Preferences:
A representative consumer who lives in a Chinese city, i, receives the following utility through the consumption of goods and services in sector $\omega \in \{0, 1, ..., \Omega\}$:

$$U_i = \sum_{\omega=0}^{\Omega} \beta_i^{\omega} \ln C^{\omega},$$

where $\sum_{\omega=0}^{\Omega} \beta_i^{\omega} = 1$ and $\beta_i^{\omega} \geq 0$.

We have a timing assumption to consider the consumer's choice problem. First, a consumer sets her budget for goods and services in each sector, and next she decides on the detailed types of products she wishes to consume. We assume one of the sectors ω denotes the index for the tourism and travel-related services sector, and we omit that indicator in the following equations. The Cobb-Douglas utility function implies that a consumer in city i spends $X_i = \beta_i Y_i$ on travel. Y_i is the aggregate income of a Chinese city i. Given this budget for travel, a consumer decides her destination and visits there to consume travel-related services.

A consumer in city i receives the following utility in destination country j:

$$\ln C_{ij} = \ln \frac{a_j q_{ij}}{\tau_{ij}},$$

where a_j is the amenity that each destination provides to a consumer, q_{ij} is the quantity of travel services and goods, and τ_{ij} is the iceberg travel costs. The quantity of consumption is $q_{ij} = X_i/p_j = \beta_i Y_i/p_j$, and p_j is the price of travel services and goods in the destination j. We restate the utility from travel:

$$\ln C_{ij} = \ln \frac{a_j \beta_i Y_i}{\tau_{ij} p_j} = \ln a_j + \ln \beta_i + \ln Y_i - \ln \tau_{ij} - \ln p_j. \tag{C.1}$$

Tourism Service Technology:
There are J foreign countries, and each country offers a different level of amenity a_j to each

traveler. We assume that a_j has a Fréchet distribution with the cumulative distribution function (CDF):

$$G_j(a) = exp(-(a/A_j)^{-\theta}),$$

where A_j is a country-specific attractiveness as a travel destination, and θ is a dispersion parameter that is common to all destinations. If a_j is distributed Fréchet, $\ln a_j$ has the Gumbel distribution (the type-I generalized extreme value distribution), and its CDF is $\hat{G}_j(\ln a) = \exp[-\exp(-\theta(\ln a - \ln A_j))]$. Assume there are N_j locations to visit in each country j. Each traveler draws her idiosyncratic preference shock for each location and decides which location she visits as the main destination in country j. The maximum of N draws from the the Gumbel distribution, $\hat{G}_j(\ln a)$, has the double exponential distribution: $\exp[-\exp(-\theta(\ln a - \ln A_j - (1/\theta)\ln N_j))]$. Using equation (C.1), the expected utility through traveling to country j from city i is:

$$E[\ln C_{ij}] = \ln A_j + (1/\theta)\ln N_j + \ln \beta_i + \ln Y_i - \ln \tau_{ij} - \ln p_j + \epsilon_{ij},$$

where ϵ_{ij} is i.i.d. with the Gumbel distribution and its CDF is $\exp(-\exp(-\theta\epsilon))$. According to Anderson et al. (1992), the choice probability takes the multinomial logit formula[35]:

$$\pi_{ij} = \frac{\exp[\ln N_j + \theta(\ln A_j + \ln \beta_i + \ln Y_i - \ln \tau_{ij} - \ln p_j)]}{\sum_{j=1}^{J} \exp[\ln N_j + \theta(\ln A_j + \ln \beta_i + \ln Y_i - \ln \tau_{ij} - \ln p_j)]}.$$

This choice probability implies that the fraction of consumers in city i who choose to travel to country j increases with the size of the origin city and the destination country, Y_i and N_j, as well as the destination's attractiveness A_j. Conversely, the probability decreases with travel costs τ_{ij}, and the price in the destination p_j.

Bilateral Flow of Traveler Spending:
The expected bilateral flow of transactions by travelers from city i to destination j is

$$X_{ij} = \pi_{ij} X_i,$$

where X_i is the total traveler spending in city i such that $X_i = \sum_{j=1}^{J} X_{ij}$. Using $X_i = \beta_i Y_i$ and adding a year subscript, t, the expected flow of traveler spending from city i to destination

[35]It is because the probability that a consumer in city i chooses j as her travel destination will converge by the law of large numbers, as the number of foreign countries, J, is sufficiently large

j in year t is

$$X_{ijt} = N_{jt} A_{jt}^{\theta} (\beta_{it} Y_{it})^{1+\theta} (\tau_{ijt} p_{jt})^{-\theta} \Phi_{it}^{\theta}, \tag{C.2}$$

where $\Phi_{it} = \left[\sum_{j=1}^{J} N_{jt} \left(\frac{\tau_{ijt} p_{jt}}{A_{jt} \beta_{it} Y_{it}} \right)^{-\theta} \right]^{-\frac{1}{\theta}}$.

Air Connectivity:

There are two types of costs for consumers to travel to their destination countries: one is time-varying—the degree of air flight connectivity between Chinese city i and foreign country j—while the other is time-invariant—characteristics that are common to i and j, such as language barriers. We can express the total trade costs, τ_{ijt}, as

$$\tau_{ijt} = D_{ijt} \, e^{\alpha_{ij}}, \tag{C.3}$$

where D_{ijt} is air flight connectivity at t, and α_{ij} is common characteristics between i and j.

Gravity Equation:

Taking logs of the variables in (C.2) and using (C.3), we obtain the equation that represents the log of the expected trade flow in traveler spending from Chinese city i to country j in year t:

$$\ln X_{ijt} = \underbrace{(1+\theta)\ln \beta_{it} + (1+\theta)\ln Y_{it} + \theta \ln \Phi_{it}}_{\text{Chinese city effects}} + \underbrace{\theta \ln A_{jt} - \theta \ln p_{jt} + \ln N_{jt}}_{\text{destination effects}} - \underbrace{\theta \ln D_{ijt} - \theta \alpha_{ij}}_{\substack{\text{city-destination} \\ \text{effects}}}. \tag{C.4}$$

This equation shows that the flow of traveler spending in year t depends on effects specific to Chinese city i, effects specific to foreign destination j, and the origin-destination effects of travel costs.

Equation (C.4) represents the expected value of transactions made by travelers from Chinese city i to foreign country j. We add an error term, ϵ_{ijt}, that captures measurement error in card transactions to (C.4), and use the resulting equation to estimate the empirical relationship between card transactions and air connectivity. We apply the inverse hyperbolic sine transformation to transaction value, X_{ijt}, and air connectivity, D_{ijt}, because of the presence of many zeros in the air connectivity, D_{ijt}. This transformation closely resembles the natural logarithm of the variables but accommodates zero values.